
SEGMENTATION AND PAGING

IN OPERATING SYSTEMS

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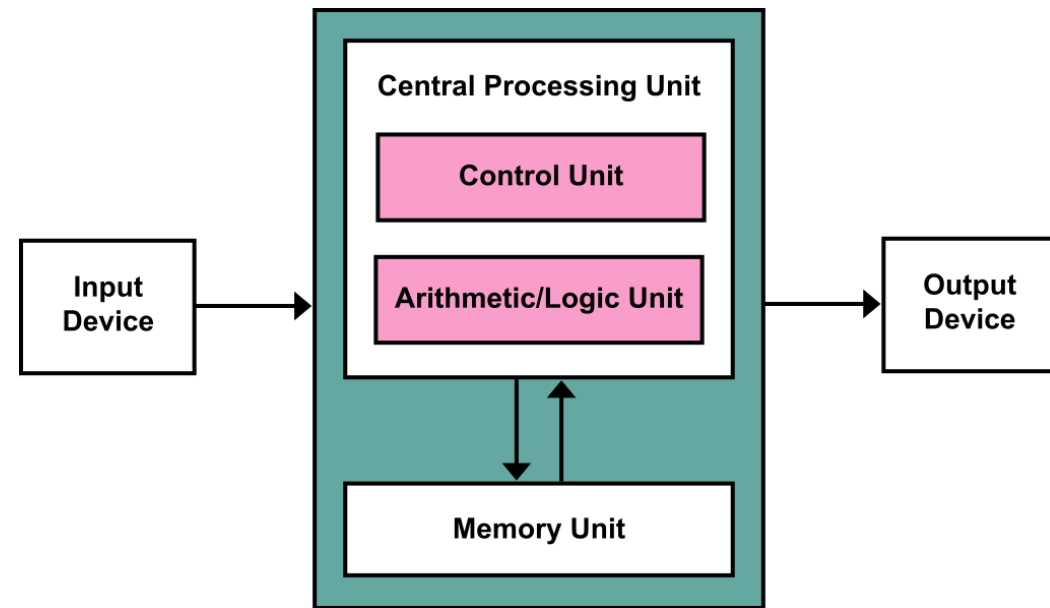
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WHAT IS MEMORY MANAGEMENT ?

Memory management is the process of controlling and coordinating the computer's main memory to ensure efficient use by programs and applications

Primary Objective:

- Efficient Memory Utilization
- Process Isolation and Protection
- Memory Allocation and Deallocation
- Multiprogramming Support
- Swapping and Virtual Memory Support
- Fast Access
- Relocation and Address Mapping



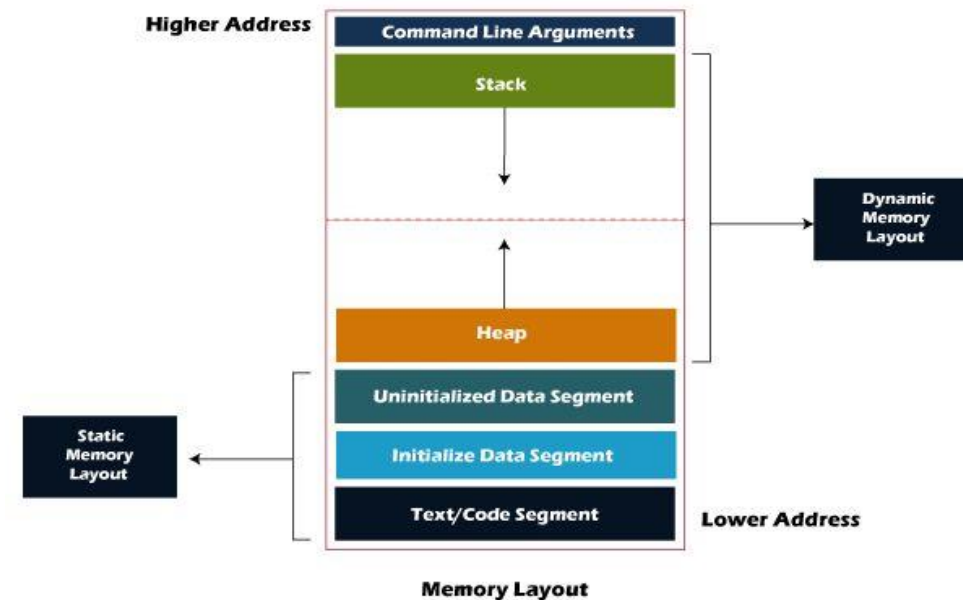
Block Diagram of Computer

SEGMENTATION OVERVIEW

Segmentation divides memory into logical units like code, data, and stack, each with a base address and limit. The OS uses these to translate logical addresses into physical ones, ensuring memory protection, modularity, and efficient access.

Advantages:

- Logical Organization
- Memory Protection
- Dynamic Growth
- Efficient Access Control
- Simplifies Sharing



HOW SEGMENTATION OPERATES ?

Logical address: <segment number, offset>.

Actions to take:

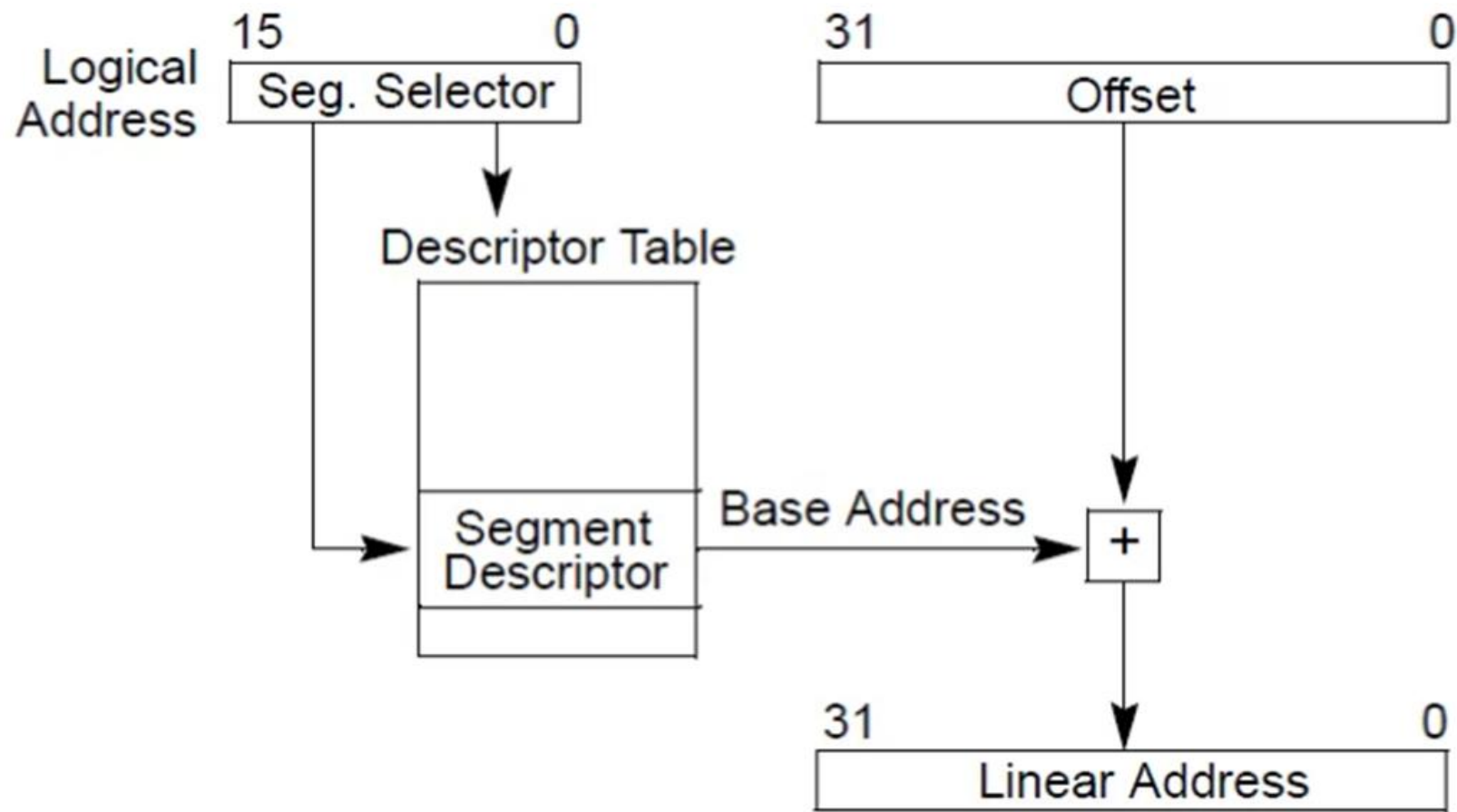
- In the segment table, find the segment number.
- Check if offset < limit.
- Base address plus offset is the physical address.

As an example:

Segment 2, offset 150.

Segment table: Base = 4000, Limit = 500.

Physical address = $4000 + 150 = 4150$.

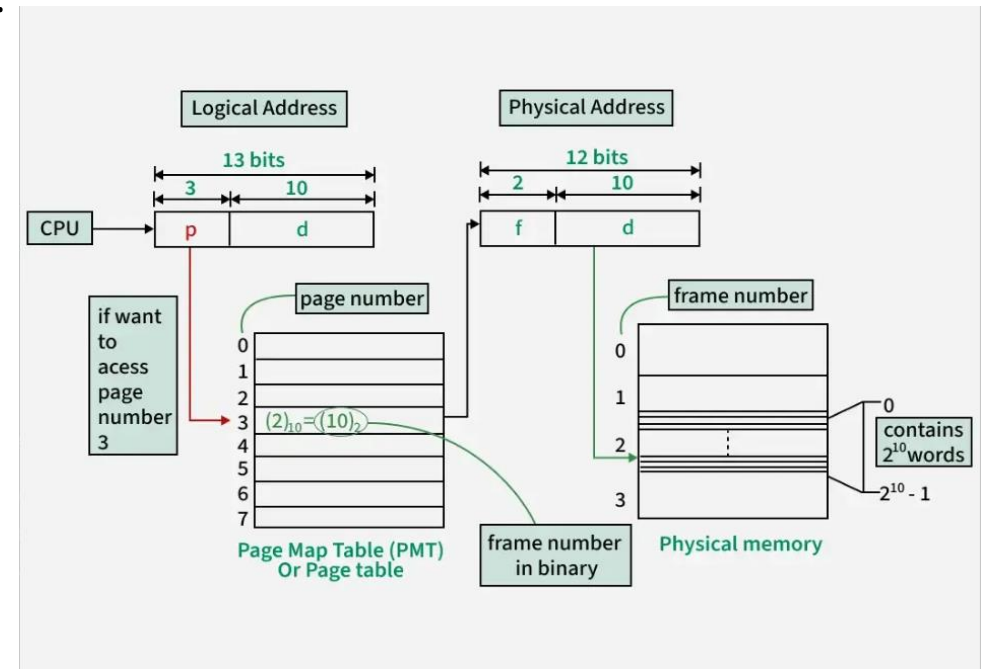


PAGING OVERVIEW

Paging is a memory management technique that divides both the logical memory and physical memory into fixed-size blocks — pages (logical) and frames (physical). When a program runs, its pages are loaded into available frames, allowing non-contiguous memory allocation.

Advantages:

- No External Fragmentation
- Efficient Memory Use
- Simplifies Swapping
- Supports Virtual Memory:
- Isolation and Protection:



OPERATION OF PAGING

Logical address: <page number, offset>.

Actions to take:

- Obtain the frame number
- Use the page table to find the page number.
- The physical address is equal to the frame number times the page size plus offset

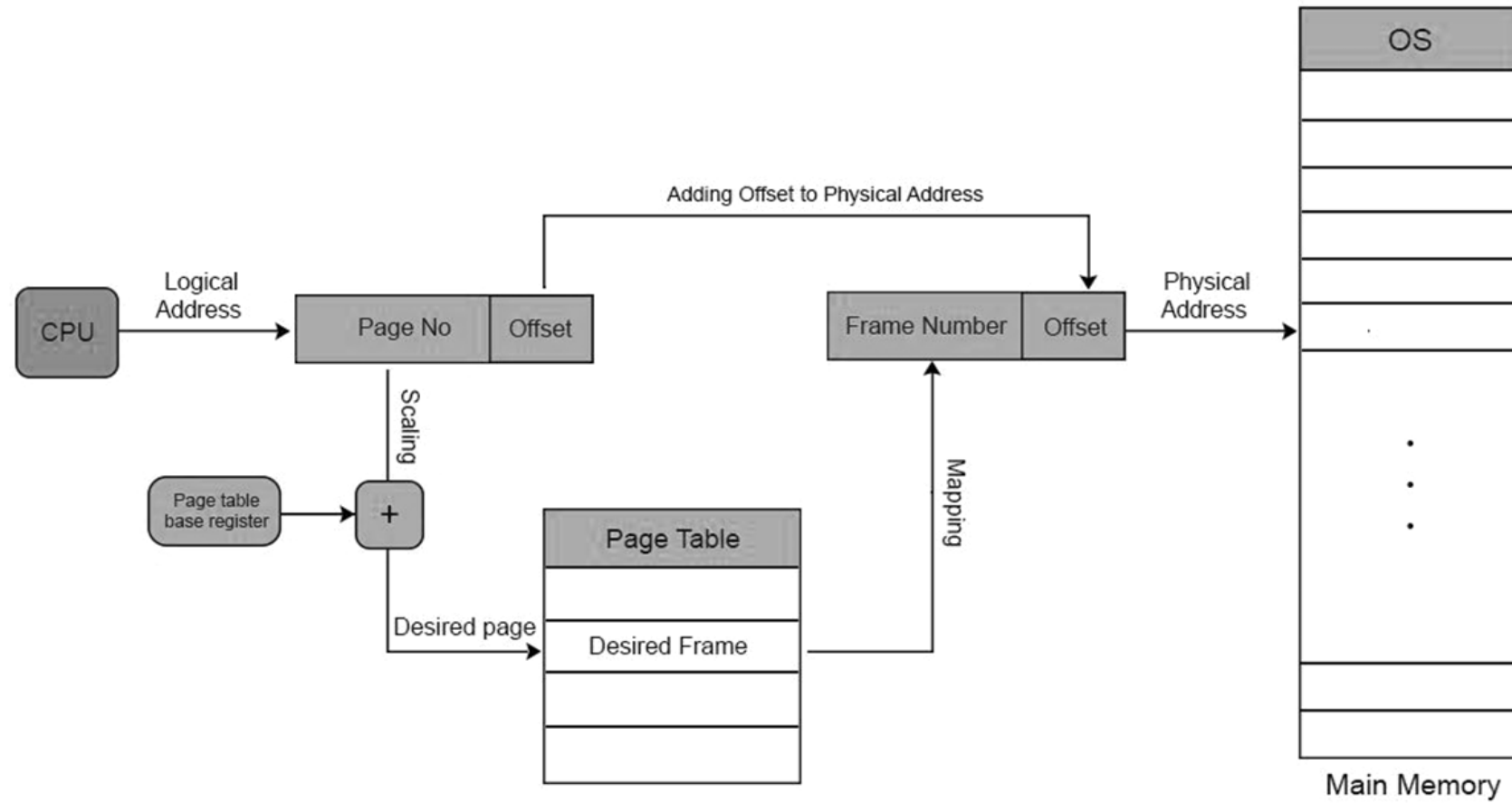
As an example:

Page size = 4KB.

Logical address: Page 3, offset 1024.

Page table: Page 3 $\rightarrow$ Frame 7.

Physical address = $7 \times 4096 + 1024 = 29696$

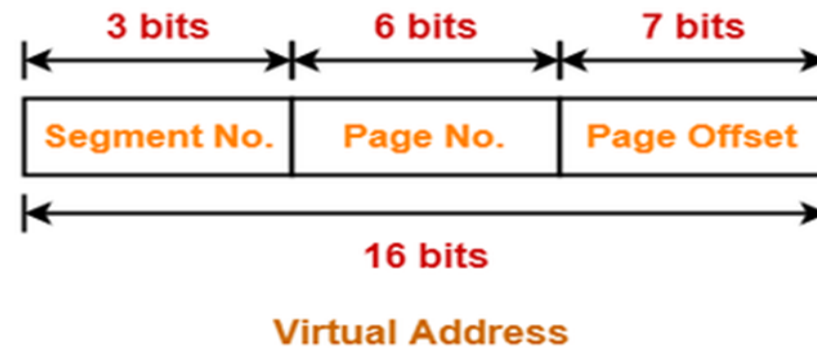


SEGMENTATION WITH PAGING

Segmentation with paging is a memory management technique that combines the logical division of segmentation with the efficient memory use of paging.

Process:

Logical address $\rightarrow$ Segment $\rightarrow$ Page $\rightarrow$ Physical address.



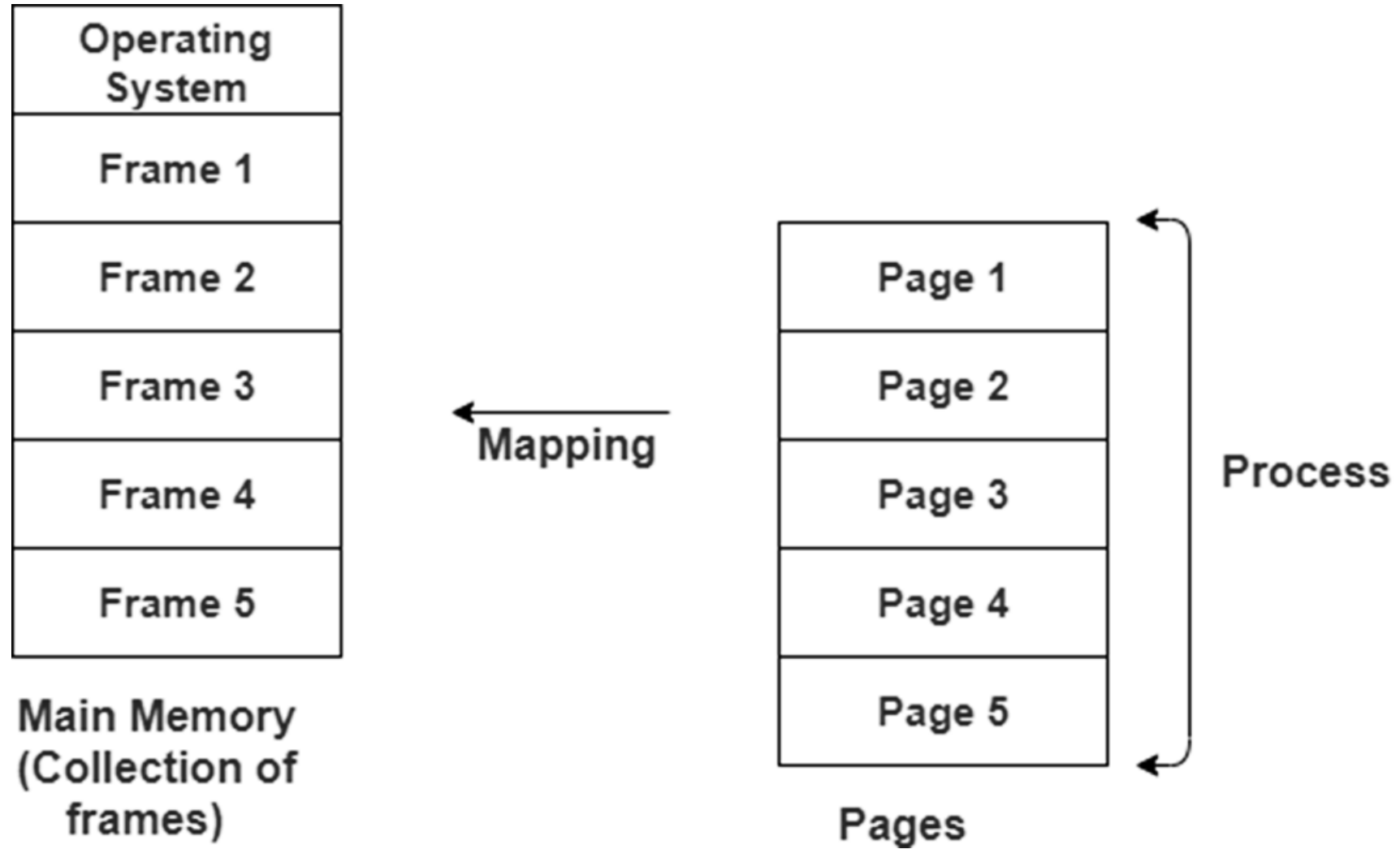
SEGMENTATION + PAGING

Logical address: Segment 1, offset 5000.

Page size = 4KB.

Steps:

- Segment table: Segment 1 $\rightarrow$ Page table.
- Offset 5000 = Page 1, offset 904 ($5000 \div 4096 = 1, 5000 \bmod 4096 = 904$).
- Page table: Page 1 $\rightarrow$ Frame 5.
- Physical address = $5 \times 4096 + 904 = 21384$.



COMPARING SEGMENTATION AND PAGING

SEGMENTATION	PAGING
Divides memory into logical segments	Divides memory into fixed-size pages
Segments are variable in size	Pages are of fixed size
Address = Segment number + Offset	Address = Page number + Offset
Preferred for logical memory management	Preferred for efficient memory allocation

WHERE ARE THEY USED ?

- Segmentation: Older systems (e.g., Intel 8086).
- Paging: Modern OS like Linux, Windows.
- Combined: x86 architecture, modern multitasking OS.

CRITICAL UNDERSTANDING

- Segmentation organizes memory logically based on program structure (e.g., code, data, stack).
- Paging simplifies memory allocation by dividing memory into fixed-size blocks.
- Both techniques support secure, efficient memory management.
- Modern operating systems often use a combination of segmentation and paging.
- Understanding address translation is essential for designing effective memory management in operating systems.